

在AI场景下使用Volcano

目录

- 01 Volcano支持AI
- 02 使用Kubeflow运行AI作业
- 03 通过Volcano Job提交AI作业

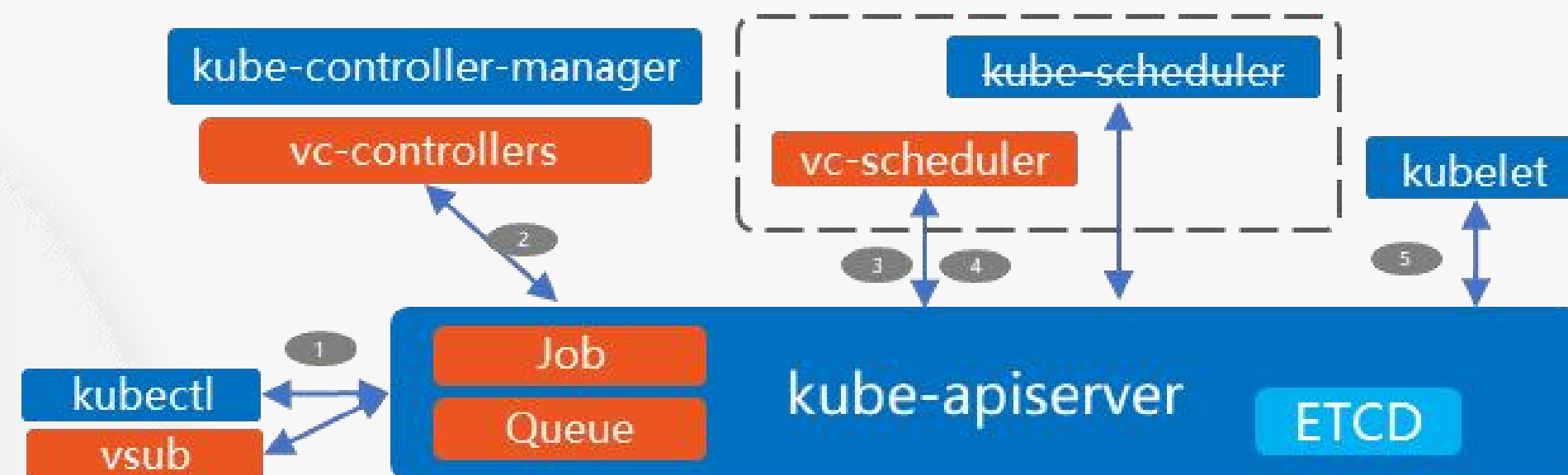


01

Volcano支持AI



Volcano支持AI



Volcano在AI场景下的关键优势

- 丰富的高阶调度策略：公平共享、弹性调度、拓扑感知、基于SLA调度等，以及抢占、回收等资源管理
- 作业生命周期管理：Pytorch, MPI, Tensorflow等批处理负载的统一生命周期管理、作业依赖、作业内任务依赖、作业TTL、细粒度作业控制等
- 资源管理增强：动态资源共享，多租户资源租借/抢占，异构资源支持等
- 高性能优化：网络、吞吐、容器引擎、kube-apiserver、etcd优化，如单调度器实例1.5k Pod/s调度吞吐

02

使用Kubeflow运行AI作业

安装 Kubeflow training-operator

```
root@ecs:~/go/src/github.com#
```



运行Pytorch作业

```
apiVersion: "kubeflow.org/v1"
```

```
kind: PyTorchJob
```

```
metadata:
```

```
  name: pytorch-sample
```

```
  namespace: kubeflow
```

```
spec:
```

```
  pytorchReplicaSpecs:
```

```
    Master:
```

```
      replicas: 1
```

```
      restartPolicy: OnFailure
```

```
      template:
```

```
        spec:
```

```
          schedulerName: volcano
```

```
          containers:
```

```
            .....
```

```
    Worker:
```

```
      replicas: 1
```

```
      restartPolicy: OnFailure
```

```
      template:
```

```
        spec:
```

```
          schedulerName: volcano
```

```
          containers:
```

```
            .....
```



作业类型: PyTorchJob



申明Volcano为调度器



申明Volcano为调度器



运行Pytorch作业

```
root@ecs- ~#
```



运行Tensorflow作业

```
apiVersion: "kubeflow.org/v1"
kind: "TFJob"
metadata:
  name: "dist-mnist-for-e2e-test"
spec:
  tfReplicaSpecs:
    PS:
      replicas: 2
      restartPolicy: Never
      template:
        spec:
          schedulerName: volcano
          containers:
            .....
  Worker:
    replicas: 4
    restartPolicy: Never
    template:
      spec:
        schedulerName: volcano
        containers:
          .....
```



作业类型: TFJob



申明Volcano为调度器



申明Volcano为调度器



03

通过Volcano Job提交AI作业

不使用operator运行AI作业

```
apiVersion: batch.volcano.sh/v1alpha1
kind: Job
metadata:
  name: tensorflow-benchmark
  labels:
    "volcano.sh/job-type": "Tensorflow"

spec:
  minAvailable: 3
  schedulerName: volcano

  plugins:
    env: []
    svc: []

  policies:
    - event: PodEvicted
      action: RestartJob
```

→ 设置作业类型

→ minAvailable声明执行作业需要启动的最小pod数

→ plugins声明在作业中生效的job级插件

→ policies声明job级事件发生时volcano job对应的行为

不使用operator运行AI作业

```
tasks:
- replicas: 2
  name: worker
  policies:
  - event: TaskCompleted
    action: CompleteJob
  template:
    spec:
      containers:
      - command:
        - sh
        - -c
        - |
          PS_HOST=$(cat /etc/volcano/ps.host
| sed 's/$/&:2222/g' | tr "\n" ",");
          WORKER_HOST=$(cat
/etc/volcano/worker.host | sed 's/$/&:2222/g' | tr
"\n" ",");

          python tf_cnn_benchmarks.py
          image: volcanosh/example-tf:0.0.1
          name: tensorflow
          ports:
          - containerPort: 2222
            name: tfjob-port
```

→ policies声明task级事件发生时
volcano job对应的行为

→ HOST声明`ps`和`worker`任务通信时
采用的host

→ ports声明ssh通信时访问的端口号

不使用operator运行AI作业

```
root@ecs-: ~#
```



加入Volcano社区



Website: <https://volcano.sh/en/>



Github: <https://github.com/volcano-sh/volcano>



Slack Channel: <https://volcano-sh.slack.com/>



Follow Us



Join Group



谢谢

